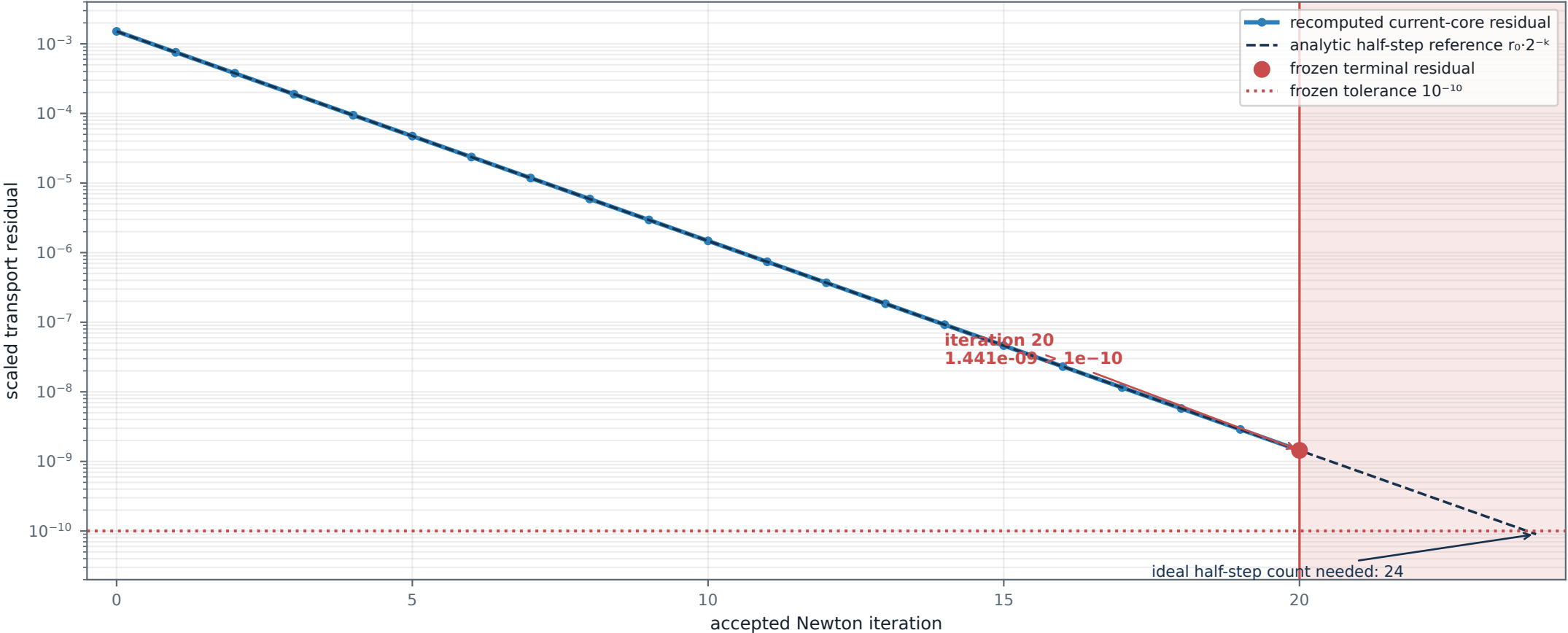
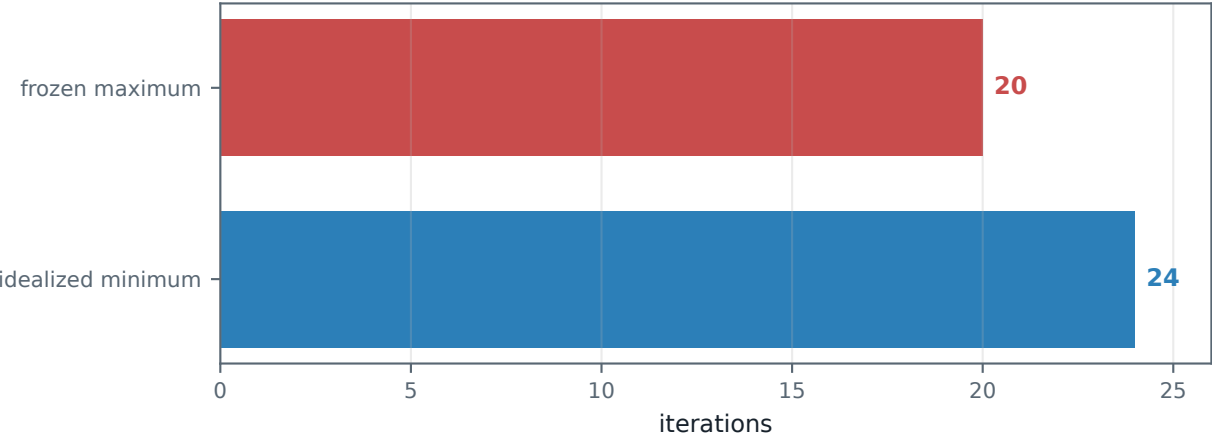


NON-SCIENTIFIC DIAGNOSTIC — minimal QN fixture, not production or oracle evidence

a Inner Newton half-step contract versus iteration budget



b Four-iteration incompatibility



c Scope and latent outer-block risk

Observed  $r_0$  / admissible  $r_0 = 14.41\times$   
Jacobian directional error =  $1.734e-10$   
Outer block: 0.5 relaxation / 12 steps /  $1e-8$   
requires initial mismatch  $\leq 4.096e-5$  (latent risk only)

Blue trace is recomputed; dashed curve is the analytic half-step reference.